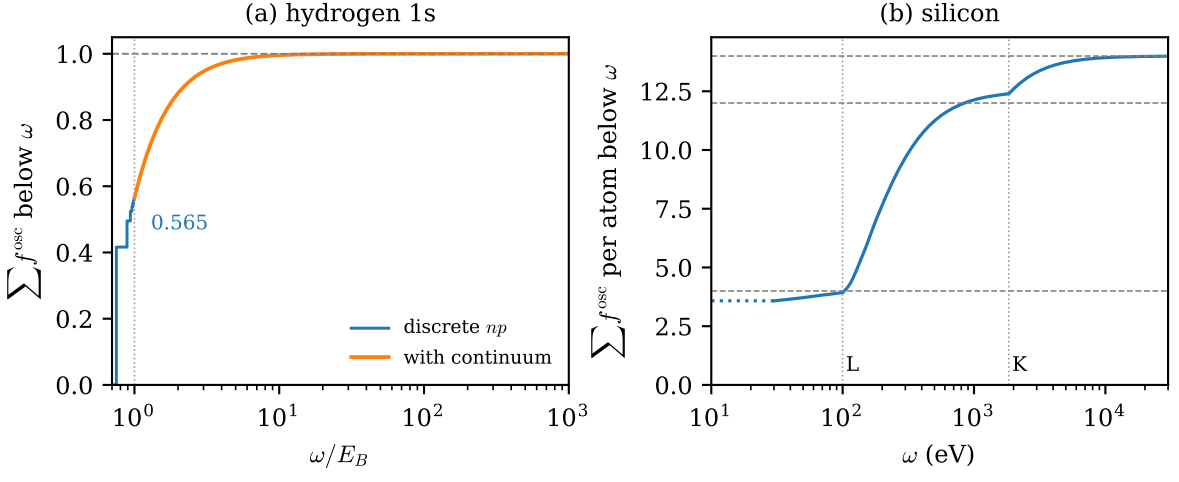
(a) hydrogen 1s
$\sum f^{\mathrm{osc}}$ below ω
1.0
0.8
0.6
0.4
0.2
0.0
0.565
discrete np
with continuum
10^0
10^1
10^2
10^3
ω/E_B
(b) silicon
$\sum f^{\mathrm{osc}}$ per atom below ω
12.5
10.0
7.5
5.0
2.5
0.0
L
K
10^1
10^2
10^3
10^4
ω (eV)